

Qatar-1b

R-0485

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180719 · created 2026-07-20T08:05:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458318.6969 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.055
Transit depth (Rp/Rs)^2	3.64 +/- 2.11 [%]
Orbital Inclination (inc)	81.25 +/- 2.88
Airmass coefficient 1 (a1)	0.44 +/- 0.21
Airmass coefficient 2 (a2)	0.56 +/- 0.85
Scatter in the residuals of the lightcurve fit is	45.03 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.042 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.055 vs 0.1463 (deviation 0.65σ)
Duration fit vs published	0.042 d vs 0.0692 d (deviation 0.75σ)
Mid-time O–C	-3.0 min
Depth SNR	2.8
Residual scatter	45.03 %

Light curve

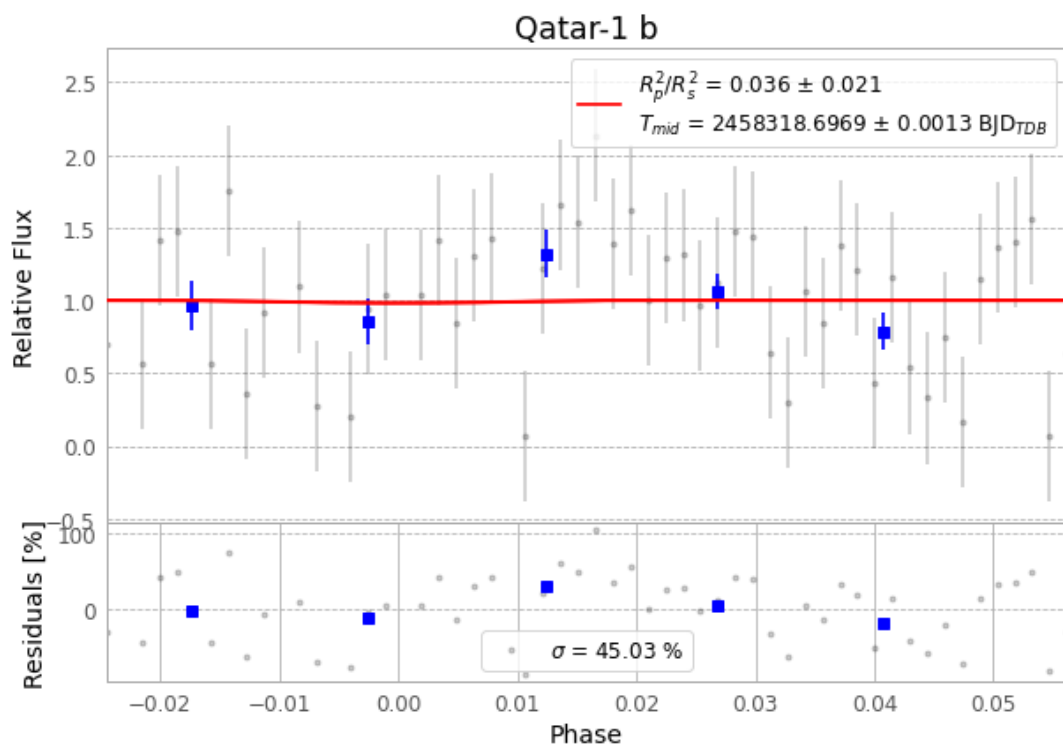


Figure 1 — FinalLightCurve_Qatar-1 b_2018-07-18.png (SHA-256 80fa49985c441377...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [104, 327], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 46fd7865fd9af2d1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

56 FITS frames (37 MB), Qatar-1180719035112.FITS → Qatar-1180719063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180719.log (9f56827d75ff...), FinalLightCurve_Qatar-1 b_2018-07-18.png (80fa49985c44...), FinalLightCurve_Qatar-1 b_2018-07-18.csv (9573c2d86a12...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 08:05Z.